

# Jason Li

Kaysville, UT — 2ljason.li@gmail.com — Website — LinkedIn — GitHub

## EDUCATION

### University of Utah

*Ph.D. in Computing, Data Management and Analysis Track*

- GPA: 4.0/4.0

Salt Lake City, UT

Aug. 2025 – Present

### University of Florida

*B.S. in Computer Science and Statistics*

- GPA: 3.94/4.0; GRE: 337 (Q: 170, V: 167)
- Honors: UF Honors Program, Undergraduate Research Scholars Program, Benacquisto Scholar (Full-Ride Scholarship), Dean's List, National Merit Finalist

Gainesville, FL

Aug. 2021 – May 2025

## PUBLICATIONS

1. **Jason Li**, Rodrigo de la Nuez Moraleda, Dhruv Meduri, Kelin Zia, and Bei Wang, “Explainable Reaction Atlas,” *IEEE Workshop on Topology meets Artificial Intelligence (TopoInVis Connect)*, *IEEE VIS*, 2026. **Accepted.**
2. Andres Espinosa, Cody Flynn, **Jason Li**, Han Mach, Maximillian Banach, Cathy Quan, Jorg Peters, and Brian Stensrud, “Multi-Agent Board Game Strategy Through Simulation,” *Interservice/Industry Training, Simulation and Education Conference (I/ITSEC)*, 2025. **Best Paper Award.**

## AWARDS

- **I/ITSEC 2025 Best Paper Award** — *Multi-Agent Board Game Strategy Through Simulation*

## RESEARCH EXPERIENCE

### Research Assistant

*Topological Intelligence Lab (TILab), University of Utah*

- Conduct research under Ph.D. advisor Professor Bei Wang Phillips at the intersection of explainable AI, deep learning, and data visualization.
- Develop methods for visualizing and interpreting latent representations learned by deep learning models to improve model transparency and understanding.
- Investigate explainable AI techniques and human-AI interaction methods for enabling users to explore, interpret, and reason about complex model behavior.

Dec. 2025 – Present

Salt Lake City, UT

### Research Assistant

*Professor Leo Duan's Research Group, University of Florida*

- Statistically analyzed and visualized travel movement trends caused by Hurricane Ian using R.
- Identified traffic-volume patterns before, during, and after Hurricane Ian.
- Parsed and cleaned census-tract-level geospatial data using Python.

Feb. 2024 – May 2024

Gainesville, FL

### Research Assistant

*Embodied Learning & Experience Lab, University of Florida*

- Developed a smartwatch application to facilitate learning in everyday life for children ages 9–12.
- Reviewed and synthesized peer-reviewed literature to inform research initiatives.

Aug. 2023 – Jan. 2024

Gainesville, FL

## PROFESSIONAL & TEACHING EXPERIENCE

### Teaching Assistant

*University of Florida*

- Tutored students in Information and Database Systems.
- Graded student submissions, administered examinations, and presented lecture material during discussion sections.

Aug. 2024 – May 2025

Gainesville, FL

### Software Engineer Intern

*Fidelity Investments*

- Developed a secure web-based funds-transfer platform for digital banking using Angular and Graph API.
- Implemented backend API services to query and format data and support front-end navigation.

Jun. 2024 – Aug. 2024

### Peer Tutor

*Academic Tutoring Center, University of Florida*

Oct. 2023 – May 2024

Gainesville, FL

- Provided one-on-one and small-group tutoring in the core programming curriculum for Computer Science and Computer Engineering students.

### IT Department Business Analyst Intern

May 2022 – Aug. 2022

#### *Supplemental Health Care*

Utah

- Automated user-permission assignment, reducing user creation time by 50%.
- Implemented a notification system supporting upskilling for more than 10,000 affiliated nurses.
- Collaborated with developers, stakeholders, project managers, and QA auditors throughout SDLC processes.

## PROJECTS

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### Explainable Reaction Atlas

2026

- Developed an LLM-based explanation framework for interpreting chemical reaction latent spaces represented as interactive tree-structured atlases.
- Implemented Neighborhood and Path Explainers to summarize local reaction regions, compare branches, and analyze transitions between user-selected reactions.
- Evaluated the framework on the Schneider 50K dataset of approximately 50,000 chemical reactions, enabling explanation-guided exploration of reaction patterns and locally inconsistent regions.

### Multi-Agent Board Game Strategy Through Simulation

2025

- Developed a custom simulation environment for the turn-based strategy game *Settlers of Catan* to support reinforcement-learning experiments.
- Implemented a Deep Q-Learning agent in PyTorch using an epsilon-greedy exploration strategy.
- Trained and evaluated the agent against a state-of-the-art benchmark, achieving a 64% win rate in one-vs-one simulations.

### Household Location Choice Model

- Modeled housing choice behavior in Salt Lake County, Utah, reducing the feature set from 150 variables to 12.
- Applied stepwise regression, LASSO, principal component analysis, random forests, variable transformations, interactions, and genetic-algorithm optimization.

### Online Shopping Application

- Built an interactive full-stack shopping application using React, Ant Design, Spring MVC, Hibernate, MySQL, and Node.js.
- Developed REST APIs for registration, search, and checkout and deployed the application using Amazon EC2 and RDS.

### Online Room Booking Site

- Developed a React and Spring Boot room-booking application with reservation, search, upload, and deletion services.
- Implemented MySQL persistence, Google Cloud Storage, Elasticsearch geospatial search, and JWT authentication.

## TECHNICAL SKILLS

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**Programming:** Python, Java, JavaScript, C++, R, MATLAB, Kotlin, SQL

**Machine Learning & Data:** PyTorch, Jupyter Notebook, MySQL, R, MATLAB, Excel

**AI & LLM Systems:** OpenAI API, OpenRouter API, Model Context Protocol (MCP), Pydantic

**Frameworks & Web:** React, Angular, Spring, Spring MVC, Spring Boot, Hibernate, Node.js, Svelte, Ant Design, HTML/CSS, Android

**Cloud & Tools:** AWS EC2, Amazon RDS, Google App Engine, Google Cloud Storage, GitHub, Elasticsearch, GraphQL, Apache Tomcat, Linux, Distributed Computing, Multithreading